Journal of Power Sources 476 (2020) 228697

Contents lists available at ScienceDirect

Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour

Butyl acrylate (BA) and ethylene carbonate (EC) electrolyte additives for low-temperature performance of lithium ion batteries

Caijian Zhu [a], Weixia Lv [a], Jun Chen [a,*], Caixia Ou [a], Qian Zhang [a], Haikuo Fu [b], Hua Wang [b], Lijue Wu [b], Shengwen Zhong [a,**]

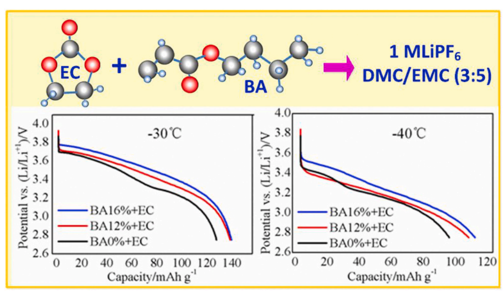
[a] School of Materials Science and Engineering, Jiangxi Provincial Key Laboratory of Power Batteries and Materials, Jiangxi University of Sciences and Technology, Ganzhou, 341000, China

[b] Guangdong Jiana Energy Technology Co Ltd, Guangzhou, 511495, China

HIGHLIGHTS

- Novel butyl acrylate (BA) with low melting point and high dielectric constant was introduced.
- The BA can effectively improve the ionic conductivity and low temperature performance.
- EC with high dielectric constant is used to further improve the ionic conductivity.
- The capacity, discharge voltage platform, cycle and rate performance are obviously improved.
- The additives of BA + EC can effectively improve the low temperature performance of LIBs.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Butyl acrylate (BA)
Lithium ion batteries (LIBs)
Electrolyte
Low temperature
Conductivity and viscosity

ABSTRACT

A novel modified reagent of butyl acrylate (BA) is introduced to improve the low-temperature performance of lithium-ion batteries (LIBs). Although the BA reagent has a slightly higher viscosity, it has a very low melting point and a very high dielectric constant, which makes the electrolyte have a high ionic conductivity, and finally the low temperature performance of lithium ion battery is obviously improved. The discharge capacities of BA0% were 116.67, 107.79, 95.65 and 74.31 mAh g^{-1} at −10 °C, −20 °C, −30 °C and −40 °C, respectively. After adding 16% BA to the electrolyte, the discharge capacities of BA16% were 127.89, 125.13, 113.28 and 91.50 mAh g^{-1} at −10 °C, −20 °C, −30 °C and −40 °C, with the increase of 9.6%, 16.1%, 18.4% and 23.1%, respectively. Moreover, ethylene carbonate (EC) is added as another modification reagent which can further improve the conductivity, increase the low-temperature capacity and achieve a better cyclic stability. As a result, the largest capacity could reach 169.12, 149.05, 138.15 and 108.94 mAh g^{-1} for the EC16% + EC10% battery, with the room-temperature capacity (identified as 180 mAh g^{-1}) retention rates of 94.0%, 82.8%, 76.8% and 60.5% at −10, −20, −30 and −40 °C respectively, which are much higher than those of BA0%, BA 8%, BA12% and 16% batteries. The mixed additives of BA + EC can not only improve the low temperature discharge

* Corresponding author.
** Corresponding author.
E-mail addresses: 741823463@qq.com (C. Zhu), 861760516@qq.com (W. Lv), chenjun@jxust.edu.cn (J. Chen), 1557907933@qq.com (C. Ou), zh_qian@126.com (Q. Zhang), fuhaikuo@jianae.com (H. Fu), 352350511@qq.com (H. Wang), lijuewu@jianae.com (L. Wu), zhongshw@126.com (S. Zhong).

https://doi.org/10.1016/j.jpowsour.2020.228697
Received 27 May 2020; Received in revised form 10 July 2020; Accepted 25 July 2020
Available online 14 August 2020